
Data Analytics Final Project Guidelines

Project Objective

The final project aims to assess your ability to perform comprehensive data analysis using Python. You are expected to demonstrate end-to-end exploratory data analysis (EDA), starting from data cleaning to deriving insights through visualizations and statistical summaries.

Project Scope

Your project will involve the following key components:

1. Data Loading and Initial Overview

- *Import the dataset using Pandas and provide an overview:*
 - Number of rows and columns
 - Data types of each column
 - Initial observations (e.g., `head()`, `info()`, `describe()`)

2. Data Pre-processing

- *Perform all necessary cleaning steps such as:*
 - Handling missing values
 - Removing duplicates
 - Correcting data types
 - Creating derived columns
 - Filtering or aggregating data

3. Exploratory Data Analysis (EDA)

- *Conduct descriptive and exploratory analysis to uncover patterns and trends:*
 - Univariate, bivariate, and multivariate analysis
 - Use groupby, pivot tables, and correlation analysis
 - Include statistical summaries to support findings

4. Visualizations

- *Use **Matplotlib** / **Seaborn** / **Plotly** to generate meaningful visualizations:*
 - Bar plots, line charts, pie charts, histograms, box plots, scatter plots, heatmaps, etc
 - Ensure visuals should have proper titles, labels, legends, and color schemes
 - Use subplots where applicable for better layout

5. Insight Generation and Report

- *Summarize key insights from your analysis:*
 - Use Markdown cells throughout the notebook to clearly explain your logic, methods, and interpretation of results
 - Highlight significant patterns, correlations, or anomalies found in the data
 - Conclude with a short summary of your overall findings and any potential recommendations or next steps
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Project Domains

Here are some sample domain ideas to help you get started.
However, you are free to choose any domain or topic of your interest.

1. Sales & E-commerce

- Sales performance dashboard
- Customer segmentation based on region or product category
- Trend and seasonality analysis of sales data

2. Finance

- Expense tracking dashboard
- Budget vs. Actual analysis
- Profitability by product or region

3. Healthcare

- Hospital/clinic patient records analysis
- Disease count trends across months/years
- Treatment cost analysis

4. Retail

- Inventory and stock analysis
- Customer purchasing patterns
- Store-wise performance comparison

5. Education

- Student performance dashboard
- Course enrollment and completion analysis
- Feedback and satisfaction trends

6. Human Resources (HR)

- Employee attrition or retention reports
- Department-wise performance or salary comparison
- Leave and attendance trends

7. Environmental

- Traffic pattern analysis across regions or time periods
- Air quality monitoring and pollutant trends
- Water consumption or waste generation tracking

8. Social Media & Public Sentiment

- Sentiment analysis of social media posts or movie reviews (e.g., Twitter, IMDB)
- Trending hashtags or topics analysis
- Public opinion analysis on a campaign or event

Dataset Requirements

The dataset should have:

- **Minimum of 500 records**
- **At least 10 features/columns** (preferably a mix of numerical and categorical data)

Data Sources

Here are some reliable sources where you can find datasets for your project:

1. **Kaggle:** [Kaggle Datasets](#)
2. **UCI Machine Learning Repository:** [UCI Repository](#)
3. **Google Dataset Search:** [Google Dataset Search](#)
4. **Data.gov (U.S.):** [Data.gov](#)
5. **Data.gov,in (India):** [Data.gov.in](#)
6. **World Health Organization (WHO):** [WHO Data](#)
7. **India Data Portal:** [India Data Portal](#)

8. **Awesome Public Datasets:** [Awesome Public Datasets](#)
 9. **OpenML:** [OpenML](#)
 10. **FiveThirtyEight:** [FiveThirtyEight Datasets](#)
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Submission Requirements

1. **Original Dataset Source**
 - The raw data file in **CSV** or **Excel** format (as obtained from the source)
 2. **Jupyter/Colab Notebook .ipynb**
 - A well-structured and well-commented notebook containing:
 - ★ **Complete Python code** covering data loading, cleaning, analysis, and visualizations
 - ★ **At least 10 different visualizations** using `matplotlib` / `seaborn` / `plotly`
 - ★ **Explanatory Markdown cells** throughout the notebook describing each step, decision, and interpretation
 - ★ **At least 5 key insights** derived from the data and explained in your own words
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Data Analytics Final Project Rubric [Total: 50 Marks]

TASK 1: Problem Definition & Dataset Selection (10 marks)

Clear problem statement, relevant domain, quality dataset (size, source, variety).

TASK 2: Data Cleaning & Pre-processing (10 marks)

Handling missing data, duplicates, formatting, derived features, and transformation.

TASK 3: EDA & Visualizations (15 marks)

Depth of analysis (univariate, bivariate, multivariate) with 10+ visuals using `matplotlib`, `seaborn`, and/or `plotly`.

TASK 4: Documentation, Insights & Presentation (15 marks)

Well-commented notebook, clear insights (at least 5), summaries, and clean presentation.
